

Learning the Time-Varying Causal Structure of S&P 500 Equities: Linear vs. Nonlinear Methods

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Causal Inference: Time Series — TUM

July 12, 2026

Abstract

We learn the causal structure among 30 high-liquidity S&P 500 stocks (2013–2026) by inferring directed graphs separately over four four-year periods, using a linear method (Granger causality) and a nonlinear method (directed information / transfer entropy), and we compare the two to assess whether linearity is an adequate assumption for financial data. We find that the choice of modeling series is decisive — log-prices yield rich, temporally evolving graphs while daily log-returns are near-white-noise outside the 2019–2022 market-stress period — that under a Gaussian assumption directed information coincides exactly with Granger causality, and that a model-free transfer-entropy estimator nevertheless recovers a largely disjoint edge set, indicating that linearity is not a complete description of the dependence.

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1 Introduction

Financial markets are a natural setting for causal discovery: the returns of different equities are highly interdependent, yet whether a given stock’s past carries genuine predictive information about another’s future — above and beyond common market movements — is far from obvious. Recovering such a directed dependence structure, and tracking how it evolves, is of interest both

for risk management (identifying which institutions transmit shocks) and as a concrete testbed for time-series causal-inference methods.

The task is to learn the causal structure among 30 high-liquidity S&P 500 stocks (six per sector across Communication Services, Energy, Financials, Information Technology, and Health Care) from their daily prices over the period 2010–2026. Because the observation window is long, we do not assume a single static graph: we divide the data into four four-year sub-periods and estimate a separate directed graph for each, so that structural change can be observed. For each period we infer the graph twice — once with a *linear* method (Granger causality, i.e. a linear structural equation model) and once with a *nonlinear* method (directed information / transfer entropy) — so that the two can be compared directly.

This design lets us ask three questions. **(i)** How does the causal structure evolve across the four periods, and does it respect the sector taxonomy? **(ii)** Is a linear model an adequate description of the dependence, or does a model-free estimator recover structure that the linear method misses? **(iii)** How sensitive are the inferred graphs to the significance threshold used to declare an edge?

To keep estimation tractable we follow the sample-complexity reductions suggested in the task: trading weeks (five market days) are treated as independent replicates, the process is assumed Markov of order $r \in \{1, 2, 3\}$, and each node is restricted to at most $K \in \{4, 5\}$ parents (Lemma 1.1). Significance is assessed by hypothesis testing (with Benjamini–Hochberg false discovery rate control) for the linear track and by thresholding for the nonlinear track, and a sensitivity analysis records how the graph changes as the threshold is varied.

Our main findings are the following. *The choice of modeling series is itself consequential*: log-prices, being strongly co-trending, yield rich and temporally evolving Granger graphs, whereas daily log-returns are close to white noise and produce sparse graphs except during the 2019–2022 market-stress period, where common movement inflates apparent connectivity. *Under the Gaussian assumption the nonlinear method adds nothing*: Gaussian directed information reduces exactly to the Granger F -test. *The genuinely model-free estimator, however, recovers a largely different edge set*: transfer entropy and Granger causality agree on only a small fraction of edges, even after matching the maximum-parents restriction, indicating that linearity is not a complete description of the dependence structure — subject to the caveats discussed in Section 5. Finally, *the Chow–Liu tree of the last period recovers the sector taxonomy* when computed on log-returns, but not on log-prices, where the common trend dominates.

2 Data and Preprocessing

Collection. Daily adjusted close prices for the 30 tickers of Table 1 of the project description were downloaded from Yahoo Finance (`code/data_collection.py`). All series were aligned to their common trading-day window; because AbbVie (ABBV) lists on 2013-01-02, the aligned panel spans **2013-01-02 to 2026-02-27** (3309 trading days, no missing values).

Transform: log-prices vs. log-returns. The pipeline is configurable (`code/period_splitter.py`). We consider two modeling series:

- **log-price** $x_t = \log P_t$ (default): the spec’s literal object, consistent with the Black–Scholes jointly-Gaussian log-price assumption; strongly cross-dependent, so it yields richer graphs. It is non-stationary (near unit root) — a caveat we return to.
- **log-return** $r_t = \log P_t - \log P_{t-1}$: stationary and statistically cleaner, but near-white-noise at the daily horizon.

Each series is split into four equal periods (by observation count) and normalised *independently within each period* (zero mean, unit variance), so no cross-period statistics leak into the estimation. We treat **log-price as the primary series**, both because it is the object named in the task

(motivated by the Black–Scholes model, in which log-prices are jointly Gaussian) and because it produces informative, non-degenerate graphs; we nonetheless report **log-return throughout as a stationary robustness comparison**, since it satisfies the independence assumptions more cleanly and provides a useful check on any conclusion that might be an artefact of the non-stationarity of price levels.

Table 1: The four estimation periods (log-price split).

Period	Date range	# obs
0	2013-01-02 – 2016-04-14	827
1	2016-04-15 – 2019-07-29	827
2	2019-07-30 – 2022-11-07	827
3	2022-11-08 – 2026-02-27	828

Independent samples. Following the project’s suggestion, trading *weeks* (5 market days) are treated as independent replicates: this is the device that yields (approximately) independent samples from a daily series. Each ticker’s period series is reshaped into a $(n_{\text{weeks}} \times 5)$ weekday matrix indexed by ISO (year, week); weeks missing any weekday (market holidays, or a partial week at a period boundary) are dropped, so that every retained week is a complete Monday–Friday observation. After this filtering each period contributes 136–142 complete weeks (140, 141, 142, and 136 for periods 0–3 respectively). Lagged regression/estimation samples are then built using *only within-week lags* — a lag never crosses a week boundary — and pooled across all weeks in the period, with each pooled sample tagged by its week of origin so that inference can, where appropriate, be made robust to within-week dependence (Section 3).

3 Methodology

3.1 Linear track: Granger causality

For an ordered pair (source X , target Y) and Markov order $p \in \{1, 2, 3\}$ we fit the pooled within-week regressions

$$\text{(full)} \quad Y_t = a_0 + \sum_{i=1}^p a_i Y_{t-i} + \sum_{i=1}^p b_i X_{t-i} + \varepsilon_t, \quad (1)$$

$$\text{(restricted)} \quad Y_t = a_0 + \sum_{i=1}^p a_i Y_{t-i} + \varepsilon_t, \quad (2)$$

and test $H_0 : b_1 = \dots = b_p = 0$ (“ X does not Granger-cause Y ”). We report two tests: the classical F -test on the pooled SSR, and a **cluster-robust Wald test** clustering the covariance by week (Cameron–Miller $F(q, G-1)$ reference, G = number of weeks), which only assumes between-week independence. p -values are corrected within each (period, order) family by Benjamini–Hochberg FDR. The headline graph uses the cluster-robust test at raw $\alpha = 0.05$ (`code/linear_granger.py`). We adopt this configuration because clustering by week is the inference consistent with the only independence we actually assume (weeks, not the days within them), while the uncorrected level keeps the graph populated enough to analyse; the more conservative FDR-corrected variant is retained as a reference. Since this choice is not unique, Section 4.4 reports how the graph density responds as α is varied across three orders of magnitude.

3.2 Nonlinear track: directed information

Gaussian directed information. Under the jointly-Gaussian assumption, the one-step causally-conditioned directed information rate is

$$\text{DI}(X \rightarrow Y) = \frac{1}{2} \log \frac{\text{Var}[Y_t \mid Y_{\text{past}}]}{\text{Var}[Y_t \mid Y_{\text{past}}, X_{\text{past}}]} = \frac{1}{2} \log \frac{\text{SSR}_{\text{restricted}}}{\text{SSR}_{\text{full}}}, \quad (3)$$

i.e. a monotone transform of the Granger statistic. Empirically its F -test p -values equal the classical Granger p -values exactly, so *under Gaussianity DI adds nothing beyond the linear test*.

Transfer entropy (model-free). The genuinely nonlinear estimator is the transfer entropy $TE(X \rightarrow Y) = I(Y_t; X_{\text{past}} | Y_{\text{past}})$, computed with the Frenzel–Pompe k -nearest-neighbour conditional-mutual-information estimator ($k = 4$, `code/directed_information.py`). Significance uses a thresholding approach calibrated by a *week-shuffled permutation null*: for each (period, order) we repeatedly permute the source series across weeks — which destroys any $X \rightarrow Y$ coupling while preserving each series’ own dynamics and the within-week block structure — and pool the resulting transfer-entropy values into an empirical null distribution. An edge is declared when its transfer entropy exceeds the $(1 - \alpha)$ quantile of that null, with $\alpha = 0.05$. The continuous transfer-entropy weights are retained so the threshold can be varied in the sensitivity analysis.

Sample-complexity reduction. We use Markov order $r \in \{1, 2, 3\}$ and restrict each node to at most $K \in \{4, 5\}$ parents (Lemma 1.1): for each target we keep the K strongest incoming edges.

4 Results

4.1 Time-varying causal graphs

Figure 1 shows the consensus graphs (an edge is retained if it is significant at ≥ 2 of the three Markov orders) for the log-price series. The structure is clearly not static: on log-prices the Granger graph is sparsest in period 1 (37 edges) and grows steadily toward the most recent period (66, 37, 85, 100 edges for periods 0–3), consistent with an increasingly interconnected market. The transfer-entropy graphs are uniformly denser than the Granger graphs (Table 2 and Figure 2); we discuss the extent to which the two methods agree on *which* edges are present in Section 4.5.

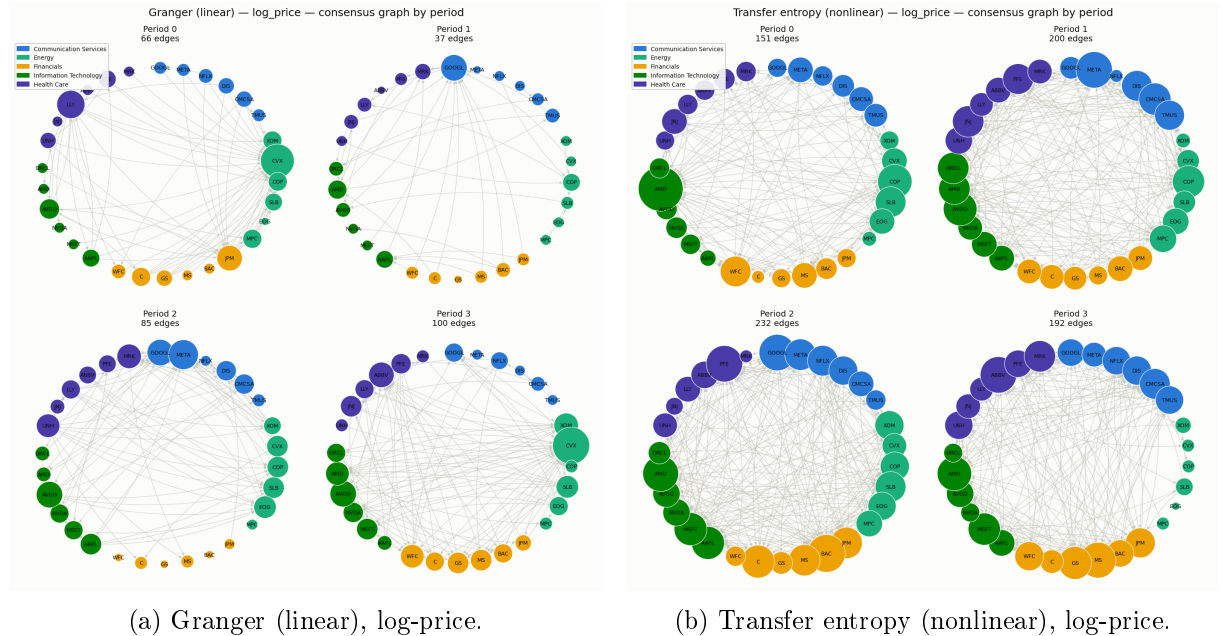


Figure 1: Estimated consensus graphs per period, nodes coloured by sector and placed in a fixed sector-grouped circular layout (identical positions in every panel), node size proportional to degree. Connectivity grows toward the most recent period, and the transfer-entropy graphs are consistently denser than the Granger graphs.

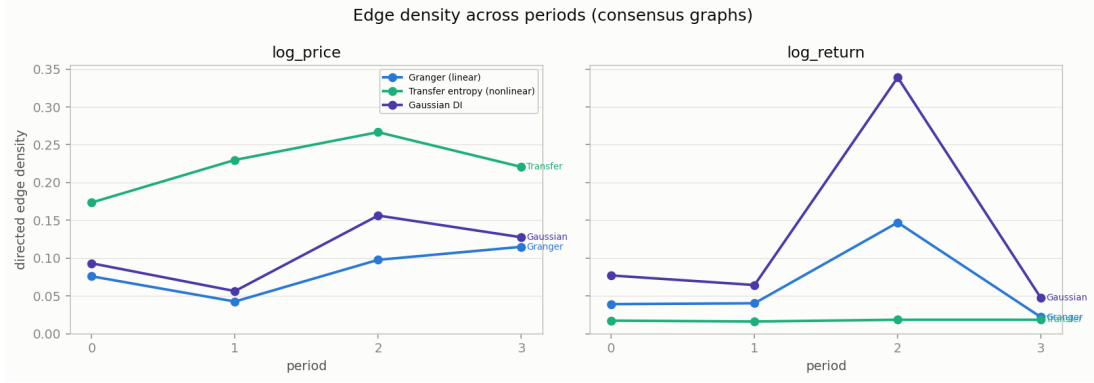


Figure 2: Directed edge density of the consensus graphs across periods, per method and mode. On log-returns the linear and Gaussian-DI density spikes sharply in period 2 (2019–2022, spanning the COVID crash and the 2022 sell-off), reflecting market-wide comovement, and is otherwise low; on log-prices the density is higher and evolves more smoothly, with transfer entropy densest throughout.

Table 2: Consensus directed edge density (edges/870).

mode	method	P0	P1	P2	P3
log-price	Granger	0.076	0.043	0.098	0.115
log-price	TE	0.174	0.230	0.267	0.221
log-return	Granger	0.039	0.040	0.147	0.022
log-return	TE	0.017	0.016	0.018	0.018

4.2 Sector structure

Partitioning each adjacency matrix into 5×5 sector blocks lets us ask whether lead-lag structure is concentrated within sectors. It is largely not: across most periods and both methods the within-sector (block-diagonal) density is comparable to the between-sector density, and the sector heatmaps (Figure 3) show no persistent strong diagonal. In other words, the detected dependence is mostly *market-wide* rather than sector-confined — with period-specific exceptions, such as Financials acting as a strong source into several sectors in the most recent log-price period. This contrasts with the tree approximation below, where a within-sector structure does emerge on log-returns.

4.3 Tree approximation of the last period

The Chow-Liu maximum-weight spanning tree of period 3 uses pairwise mutual information as edge weights (Gaussian MI and, as a nonlinear variant, KSG MI; `code/tree_approximation.py`).

Table 3: Within- vs. between-sector edges of the 29-edge Chow-Liu tree (period 3).

mode	MI estimator	within-sector	cross-sector
log-return	Gaussian	23	6
log-return	KSG	22	7
log-price	Gaussian	12	17
log-price	KSG	13	16

On log-returns, 23 of the 29 tree edges (Gaussian MI) lie within a sector, so each stock’s

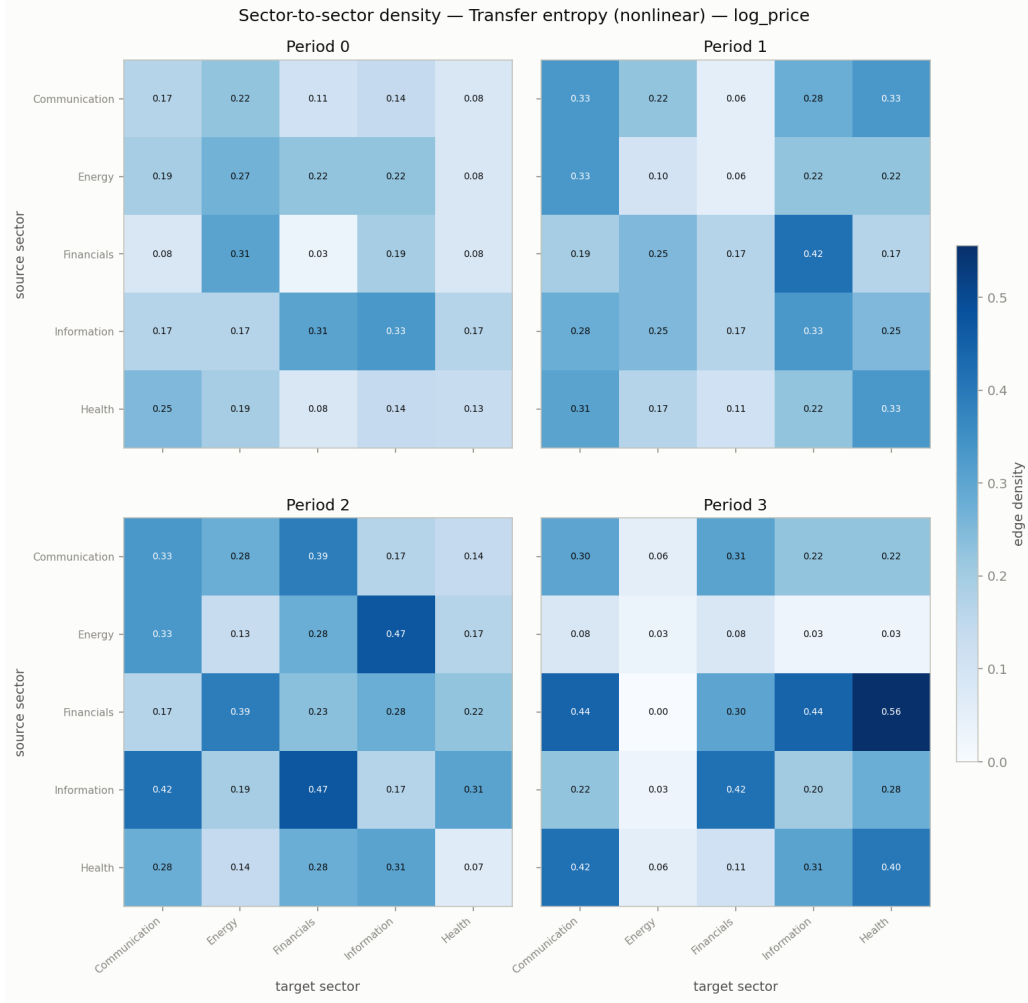


Figure 3: Source→target sector density per period (transfer entropy, log-price; rows = source sector, columns = target sector). The absence of a consistently dominant diagonal indicates that within-sector density is not systematically higher than between-sector density.

single strongest mutual-information partner is almost always a sector peer — the tree essentially reconstructs the sector grouping, and the model-free KSG variant gives the same picture (22/29). On log-prices the ordering reverses (only 12–13 within-sector edges): the shared stochastic trend makes cross-sector correlations comparable to within-sector ones, so the maximum-weight tree no longer aligns with the sectors. Log-returns are therefore the more informative series for the tree approximation, even though log-prices give the richer directed graphs — a useful illustration that the “best” series depends on the question asked.

4.4 Sensitivity analysis

Because density has no flat region in the threshold, no single cutoff is canonical; we therefore report conclusions that survive the whole sweep (the method agreement between Granger and Gaussian DI, and the period ordering) rather than any single graph.

4.5 Linear vs. nonlinear comparison

For each period we compare the headline Granger and transfer-entropy consensus graphs by their directed-edge overlap (Figure 6, Table 4). Because transfer entropy is the denser of the two, we also compare the graphs after applying the *same* $K=5$ parent cap to both, which largely

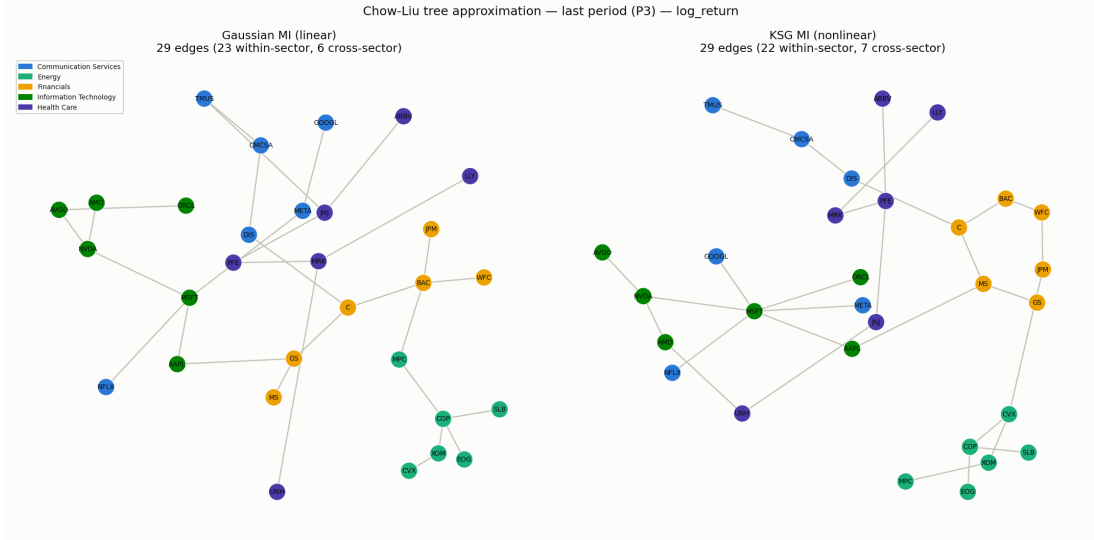


Figure 4: Chow-Liu tree, last period, log-return (node colour = sector). The tree groups stocks largely by sector — distinct Energy, Financials, Information-Technology and Communication-Services clusters are visible — so it approximately recovers the sector taxonomy.

equalises their sizes.

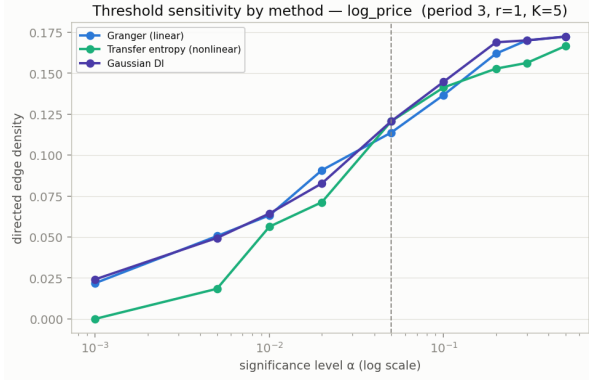
Table 4: Granger vs. transfer-entropy edge overlap (mean over periods).

comparison	log-price Jaccard	log-return Jaccard	TE edges in Granger
consensus (headline)	0.07	0.01	6–9%
matched $K=5$ (both tracks)	0.04	0.01	~5%

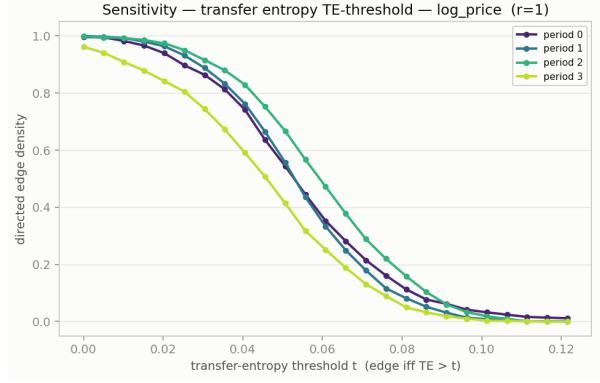
Linearity verdict. Two facts frame the answer. First, under the Gaussian assumption directed information is *identical* to Granger causality (its F -test p -values match the classical Granger p -values to machine precision), so assuming Gaussianity buys no nonlinear power whatsoever. Second, the genuinely model-free transfer-entropy graph overlaps the Granger graph only weakly — mean Jaccard 0.07 on log-prices and 0.01 on log-returns, with only ~5–9 % of transfer-entropy edges also found by Granger — and this near-disjointness *persists* when both graphs are capped at five parents per node (Table 4), so it is not merely an artefact of transfer entropy detecting more edges. We therefore conclude that *linearity is not a complete description* of the dependence structure: a model-free estimator identifies directed relationships that the linear model does not. This verdict should be read together with the caveats of Section 5, which temper how strongly it can be stated.

5 Discussion and Limitations

- **Pairwise vs. multivariate.** All edges are inferred from *bivariate* tests, which cannot distinguish a genuine $X \rightarrow Y$ influence from both stocks reacting to a common market factor; such a factor appears as many spurious pairwise edges, most visibly in the log-return period-2 spike. A natural remedy is to regress out a market proxy (e.g. the S&P 500 index) before testing, or to move to a conditional/multivariate formulation; we leave this to future work, and read the dense periods as “detected comovement” rather than idiosyncratic causation.
- **Non-stationarity of log-prices.** Log-price levels are close to a unit root, and condi-



(a) Density vs. α by method (period 3, $r=1$, $K=5$).



(b) Density vs. transfer-entropy threshold.

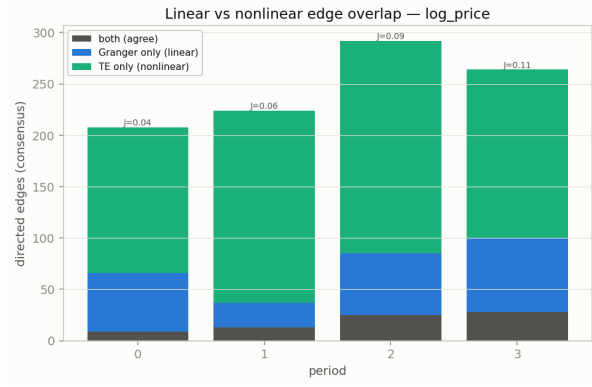
Figure 5: Threshold sensitivity. Graph density varies smoothly and monotonically with the threshold, with no wide plateau, so the absolute edge count is threshold-dependent; under the $K=5$ cap it saturates at $5 \times 30/870 \approx 0.17$. The qualitative conclusions are nonetheless robust: Granger and Gaussian DI track each other at every α , and period 2 is the densest at essentially every threshold.

tioning nonparametrically on only a few own-lags removes the common stochastic trend imperfectly. Some transfer-entropy edges on levels may therefore reflect residual shared-trend dependence rather than genuine directed coupling, which partly inflates the log-price transfer-entropy density; the stationary log-return series is the cleaner basis for the non-linearity verdict.

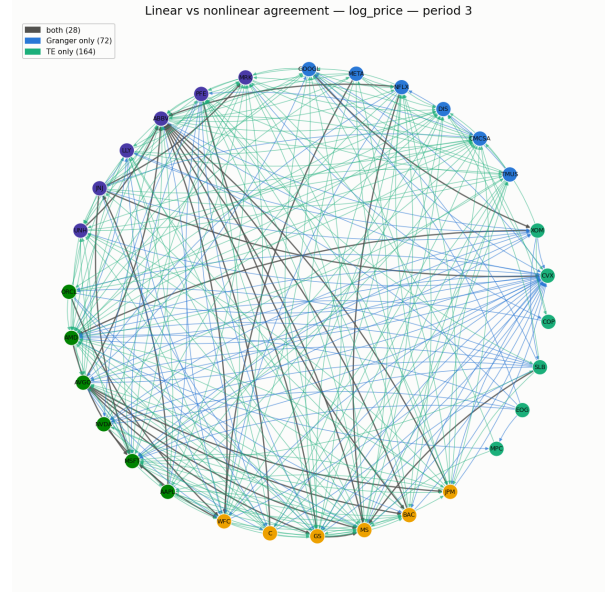
- **Power on log-returns.** Conversely, the log-return graphs are very sparse, so the near-zero linear/nonlinear overlap there may partly reflect low statistical power (few true edges detected by either method) rather than the two methods genuinely disagreeing. The stronger evidence for the linearity verdict comes from log-prices, where both graphs are populated.
- **Significance calibration.** The two tracks are not thresholded on a common scale — Granger uses a cluster-robust p -value, transfer entropy a permutation-null threshold — so part of the low overlap could stem from mismatched operating points. The sensitivity analysis (Section 4.4) mitigates this by showing the qualitative conclusions hold across a wide threshold range, but a fully matched calibration (e.g. per-pair permutation p -values for both) would make the comparison tighter.

6 Conclusion

We estimated time-varying causal graphs among 30 S&P 500 stocks with both a linear (Granger) and a nonlinear (directed-information) method, across four four-year periods and two modeling series. The inferred structure is not static: graph density evolves across periods, rising toward the present on log-prices and spiking during the 2019–2022 stress period on log-returns. Lead-lag structure is largely market-wide rather than sector-confined, yet the Chow–Liu tree of the last period cleanly recovers the sector taxonomy on log-returns — so sector structure is present in the strongest pairwise dependencies even when it does not dominate the full graph. On the central question, we found that a Gaussian assumption reduces directed information exactly to Granger causality, whereas a model-free transfer-entropy estimator recovers a largely disjoint edge set that survives a matched parent cap, indicating that linearity is not a complete description of the dependence — a conclusion to be weighed against the pairwise, non-stationarity, power, and calibration caveats above. Natural next steps are to control for a market factor (or move to conditional/multivariate estimators) to curb pairwise over-detection, and to put both tracks on



(a) Edge overlap per period (log-price).



(b) Agreement graph, period 3 (log-price).

Figure 6: Granger vs. transfer entropy: shared, linear-only, and nonlinear-only edges (`code/compare_graphs.py`).

a common footing via per-pair permutation tests, so that the linear-versus-nonlinear comparison rests on matched significance.

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